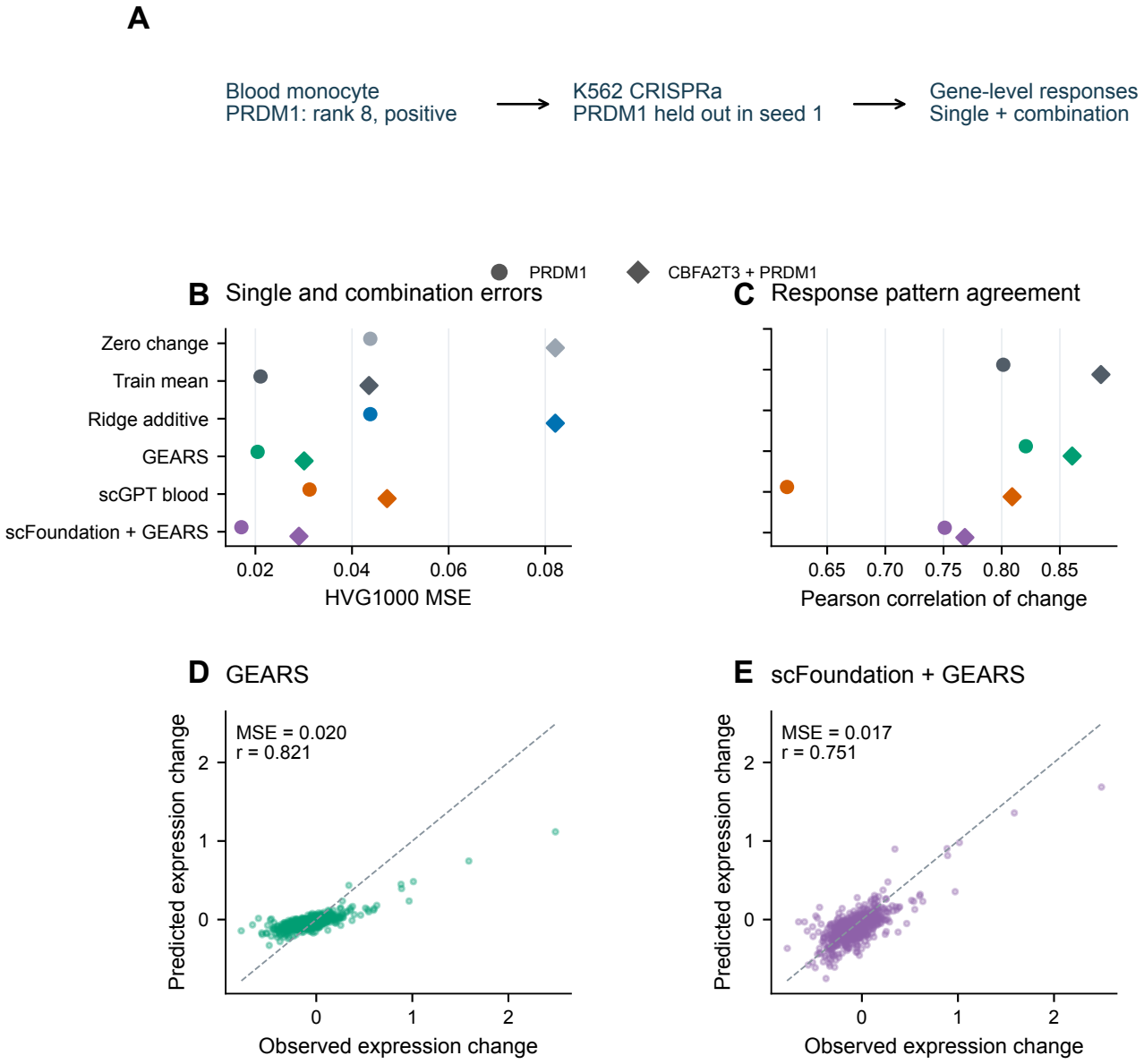

A
Blood monocyte
PRDM1: rank 8, positive
K562 CRISPRa
PRDM1 held out in seed 1
Gene-level responses
Single + combination
PRDM1
CBFA2T3 + PRDM1
B Single and combination errors
Zero change
Train mean
Ridge additive
GEARS
scGPT blood
scFoundation + GEARS
0.02
0.04
0.06
0.08
HVG1000 MSE
C Response pattern agreement
0.65
0.70
0.75
0.80
0.85
Pearson correlation of change
D GEARS
MSE = 0.020
r = 0.821
Predicted expression change
Observed expression change
E scFoundation + GEARS
MSE = 0.017
r = 0.751
Predicted expression change
Observed expression change